

5(a)(i)	833.3 ;; $\frac{280 - 30}{30} \times 100$ <i>if answer incorrect award one mark for working</i>	2
5(a)(ii)	any four from: 1 insulin secreted by, pancreas / beta cells / Islets of Langerhans ; 2 binds to receptors on (cell surface membrane of), liver / muscle, cells ; 3 glucose transporter (proteins) added to muscle cell (surface membrane) or increases permeability of muscle cell (to glucose) ; 4 more glucose enters (cells) ; 5 increase in respiration of glucose ; 6 glycogenesis ; A described	4
5(b)(i)	glucose oxidase ; peroxidase ;	2

Question	Answer	Marks
5(b)(ii)	any three from: <i>identical to human insulin</i> 1 no, allergy / immune response / antibody response / side effects / rejection ; 2 acts rapidly ; 3 do not, develop tolerance to it / need larger doses ; 4 no risk of, disease / infection ; 5 no / fewer, (named) ethical issues ;	3